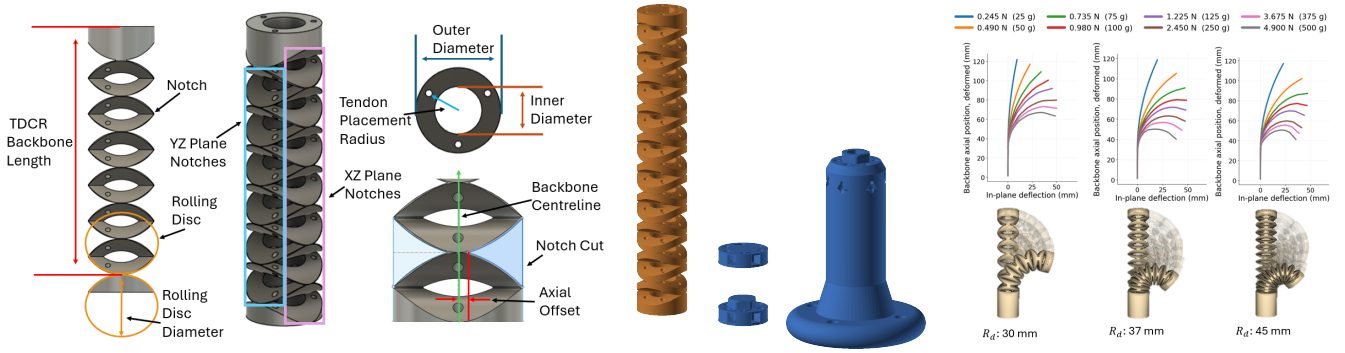
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From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

Om Mohan Pawar (25110218) Supervisor: **Prof. Madhu Vadali**
IITGN Robotics Laboratory, Department of Mechanical Engineering, Indian Institute of Technology Gandhinagar
Code, data and this report: github.com/om-mpawar/ME299-25110218

Abstract. Tendon-driven continuum robots bend along their whole length, which suits confined spaces but makes them hard to design and to model. This report describes a semester of work on a printed robot whose backbone is a TPU tube with rolling-disc notches. The first part, submitted to ICRA 2027 with me as second author, studies how the notch geometry sets the bend. An automated pipeline from CAD to finite-element simulation ran 616 simulations and three printed prototypes were tested; the disc diameter controls curvature, the length controls reach, and close to half of the tendon stroke is lost to axial compression. The second part, which I lead and which is being prepared for IEEE RA-L, is a digital twin that predicts the full shape of a modular robot from the servo commands alone. A beam model built from the CAD, with tendons that can only pull, is identified once on one and two modules. Without refitting, it predicts four-module stacks and new mountings to within 0.5 mm of FEM in 1 to 4 ms per pose. The report also covers the reading and the mistakes that shaped the work, and the plan for testing it on hardware.



(a) Notched backbone and its design parameters **(b)** Parts for the build **(c)** FEM, three disc sizes

Figure 1. The robot. (a) A TPU tube with two orthogonal families of rolling-disc notches and tendon channels in the wall. (b) Parts I designed for the four-module build: the TPU module (orange), the PLA coupler that joins two modules, the tip coupler and the base column (blue); they assemble without screws or glue. (c) SOFA simulations of the same load on three disc sizes: a larger disc gives more, shallower notches and a tighter bend.

1 Introduction

A tendon-driven continuum robot (TDCR) has no discrete joints. Tendons run through its wall, and pulling one bends the body along its length [1, 2]. Our robot is printed in TPU. Its backbone is a tube with two families of notches cut at right angles to each other, each notch shaped by a rolling disc so that neighbouring segments roll on one another instead of pinching (Fig. 1a). Modules 160 mm long can be stacked, and the tendons of distal modules pass through the proximal ones.

Two questions organised the semester. The first concerns design: which geometry gives a required bend, and what tendon stroke and force does that bend cost? The second concerns modelling: can the whole shape of a stack be predicted in real time from the servo commands, with no sensor on the robot, for any number of modules and any mounting? Section 3 summarises the answer to the first question, now under review at ICRA 2027 (D. Yadav, O. M. Pawar, Krishnapriya R, S. Gandhi and M. Vadali). Section 4 describes the digital twin I built for the second, which is in preparation for IEEE RA-L. Section 2 records what I read and how the approach changed along the way, and Section 5 sets out how the project continues.

2 What I learnt

2.1 Reading

I started from the constant-curvature model [1], which treats each section as a circular arc. It was my first baseline, and on a loaded notched robot it is far off: in Part A its error grows about five-fold as notches are added, and on our modules it misses the tip by 11 to 68 mm. The mechanics of tendon actuation come from Camarillo et al. [3,4]. Each tendon applies a moment and an axial compression to every slice it passes, the tendons of distal sections load the proximal ones, and a tendon that would have to push goes slack instead. Rucker and Webster [5] and Oliver-Butler et al. [6] give the same physics for a Cosserat rod, and I later used an independent Cosserat solver to check my own. Rao et al. [7] settled how I report errors, as a percentage of robot length. Table 1 lists what each group of papers changed in the project; the full notes on about thirty papers are in the repository.

Table 1. Reading that changed the project.

Topic	Effect on this work
Constant curvature [1, 2]	First baseline, dropped once its error on notched modules was measured
Tendon mechanics [3, 4]	Basis of the tendon map; also showed that the pull-only model was already published
Cosserat rods [5, 6]	Independent solver used to check mine
Benchmarking [7]	Errors in % of length; the 2 % hardware target
FEM for soft robots [8–10]	Ground truth; led me to find cables that could push
Shape estimation [11–13]	Defined the gap: no sensor-free transfer to longer stacks
Modular twins [14]	Closest prior work on composing modules
Physics-informed learning [15–18]	Learn the module law inside a solver

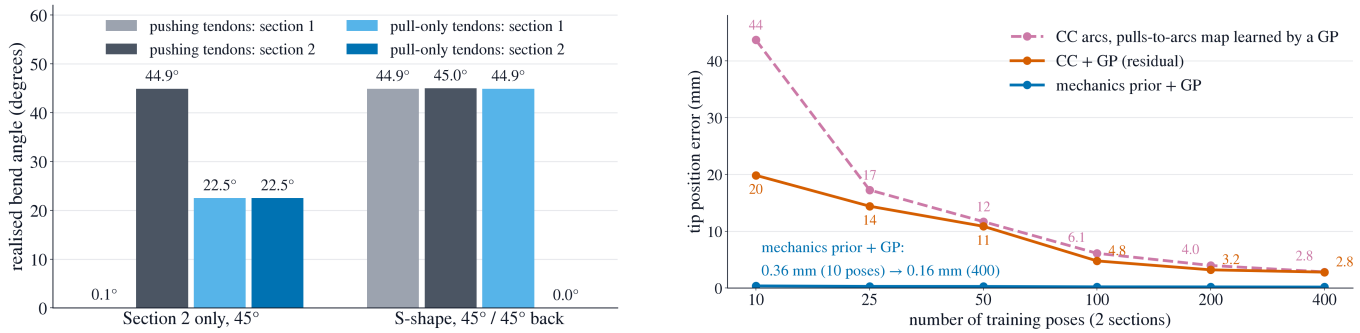


Figure 2. Two early results on the first three-tendon robot. (a) With cables that could push, commanding section 2 left section 1 straight; with pull-only cables the bend splits between the sections. (b) Tip error against training poses: a physics prior needs far less data than constant curvature.

2.2 How the approach changed

The model changed several times. It began as a virtual sensor: a constant-curvature estimate of the tip, corrected by a Gaussian process trained on FEM. The Gaussian process was then extended to predict the whole backbone through a few shape modes, which cut the tip error of a single section from 14.7 to 0.16 mm.

Two findings changed the direction. The cables in our SOFA scenes could push as well as pull, and gravity was a thousand times too weak. Once the cables were made pull-only, commanding the second section of a two-section robot also bent the first, because its tendons run through it (Fig. 2a). A pull-only model with coupling between sections captured this and cut the data the learning step needed (Fig. 2b). A literature search then showed that the same model had been published by Camarillo et al. [4]. I checked that their formulation and mine give the same configuration to 3×10^{-6} and moved my contribution to identifying the model from data and showing that it transfers. Residuals learned on two modules then failed on four, which led to learning the module law itself (Section 4).

2.3 Skills

Over the semester I learnt continuum-robot mechanics through beam and rod theory, finite-element simulation in SOFA with CadQuery and CGAL, convex optimisation with unilateral constraints, parameter identification, Gaussian-process regression and physics-informed neural networks. The habits mattered as much: testing a simulator on simple cases before trusting it, searching the literature before claiming anything is new, fixing a pass criterion before collecting data, and writing code so that every number in a paper can be regenerated by one command, with 132 automated tests behind it.

3 Geometry and compliance (ICRA 2027, under review)

SWEEP (Fig. 3) turns a set of geometric parameters into a simulated bend without manual steps. CadQuery builds the backbone with a single Boolean cut, CGAL meshes it into tetrahedra, and SOFA pulls one tendon at a time through a load sequence while tracking the centreline. Each notch is removed from a tube of outer radius R_o by a disc of diameter $D_{\mathrm{RD}} = 2R_d$, with a clearance c between cuts, which fixes the pitch

$$p = D_{\mathrm{RD}} - 2\sqrt{R_d^2 - R_o^2} + c. \tag{1}$$

The disc diameter therefore decides how many notches fit into a length L. The study covered 14 designs, each actuated through 4 tendons at 11 loads: 616 simulations, of which 614 completed. The TPU modulus was measured at 56 MPa and gravity was included. Above 0.9 N the simulations stop being reliable because of settling time-outs and self-contact, so all conclusions hold for $T \leq 0.9$ N.

The parameters separate cleanly (Fig. 3). Raising D_{RD} from 26 to 50 mm increases the notch count from 4 to 12 and the bend at 0.9 N from 54° to 143°, while the departure from a circular arc grows about five-fold. Changing the length from 100 to 200 mm nearly doubles the tip travel, from 78 to 157 mm, and changes curvature by less than 20 %.

The tendon stroke splits into a bending part and an axial part,

$$\Delta\ell = \Theta\, d + \Delta L, \tag{2}$$

and this budget closes to 0.99 on notched designs against 0.72 on a plain tube. Close to half the stroke goes into shortening the body, 14 to 17 mm of a 33 to 36 mm pull (Fig. 5). A larger disc halves the tension needed for a 110° bend, from 0.87 to 0.45 N. Because the notch families lie in two orthogonal planes, four tendons suit the structure better than three. The spread of curvature across tendon directions, $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$, falls from 6.2 % to 2.6 % in 13 of the 14 designs (Fig. 6), and two antagonistic pairs need only two motors per section.

Three prototypes were printed and loaded through a single tendon with hanging weights while ten markers were tracked (Fig. 7). The centreline agrees with simulation to within 10 to 19 % of length. The tip agrees less well, which we attribute to friction and self-contact that the model leaves out.

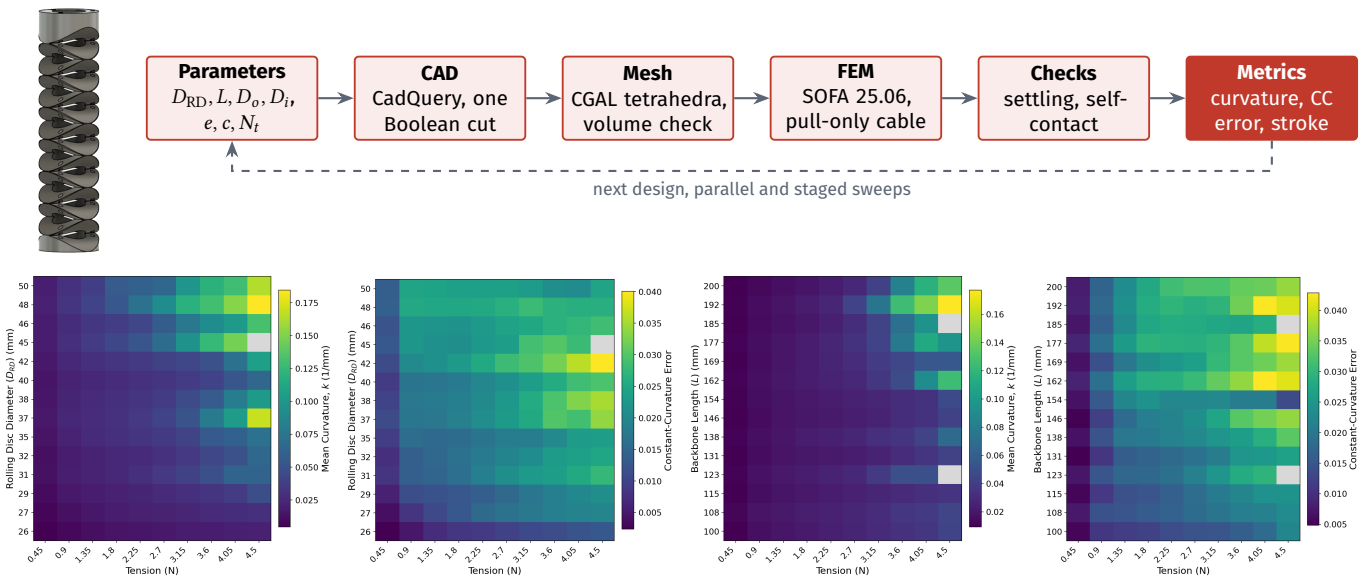
My part in this work was on the simulation side. I ported the pipeline from SOFA 23 to 25.06, which changed the batch runner, removed several components and exposed a duplicated mass. I added parallel and staged sweeps, a check of mesh volume against CAD volume and a per-run check for time-outs and self-contact, and I reported the settling and mesh-convergence limits behind the 0.9 N bound.

4 A sensor-free digital twin (for IEEE RA-L)

The twin takes the tendon displacements commanded by the servos, together with the way the robot is mounted, and returns the pose of the backbone every 5 mm. Nothing is measured on the robot while it runs.

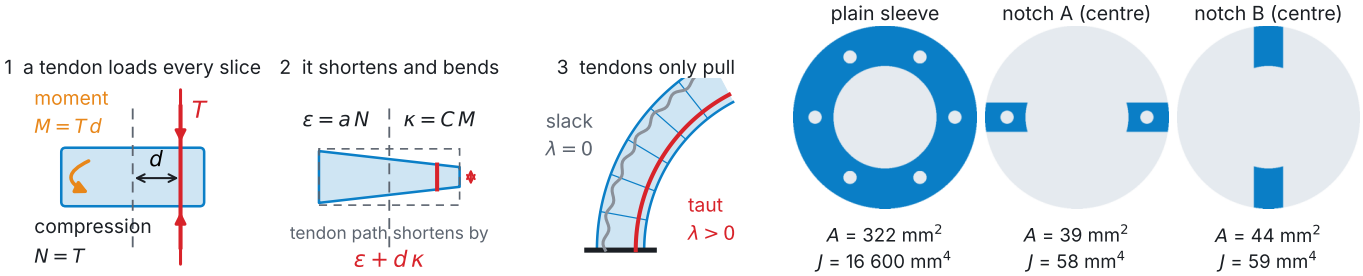
4.1 Model

The backbone is treated as a beam whose cross-section follows the CAD. A notch leaves two bridges 4 mm wide, so its area is



(a) Curvature against D_{RD} **(b)** CC error against D_{RD} **(c)** Curvature against L **(d)** CC error against L

Figure 3. Part A. Top: SWEEP, the geometry-to-simulation pipeline, with the CAD of one design and its tendon path on the left; the volume, settling and self-contact checks and the parallel sweeps are my additions. Bottom: mean curvature and constant-curvature (CC) error against tension for the disc-diameter and length sweeps. The disc diameter moves both; the length leaves curvature nearly unchanged.



(a) Loads on a slice, its deformation, and slack tendons. **(b)** Cross-sections taken from the CAD.

Figure 4. The physics of the twin. A tendon at offset d applies a moment Td and a compression T to every slice it passes; the slice shortens and bends, and tendons that would have to push go slack. In a notch only two 4 mm bridges remain, so the notches carry almost all of the bending.

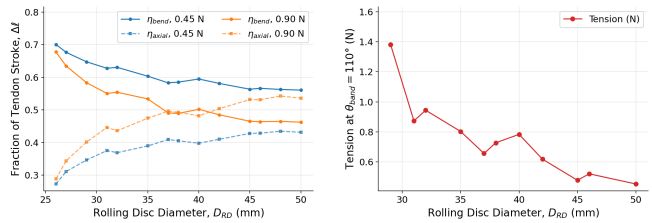


Figure 5. Stroke budget against disc diameter. Left: the bending and compression fractions of the stroke at 0.45 and 0.9 N. Right: the tension needed for a 110° bend.

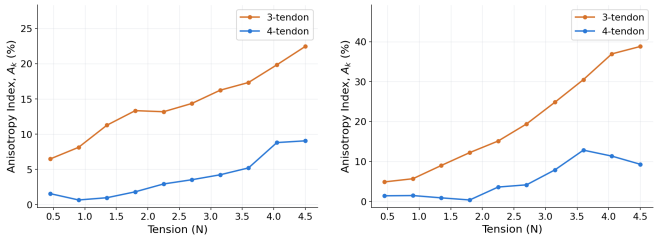
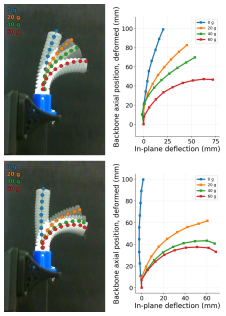


Figure 6. Anisotropy index for three and four tendons, D_{RD} = 27 mm (left) and 40 mm (right).

an eighth of that of the plain sleeve and its bending stiffness about 1/280 of it (Fig. 4b). Integrating the compliance of each slice c along the module gives the map from tendon tensions



D_{RD} (mm)	NRMSE (% of length) centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Figure 7. Printed prototypes under load and their error against simulation.

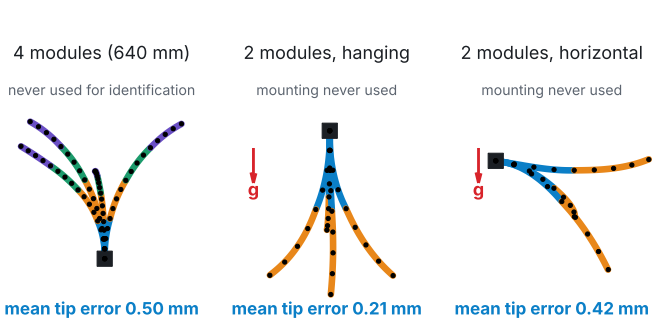
$\boldsymbol{\lambda}$ to tendon shortening,

$$\delta\boldsymbol{\ell} = G\boldsymbol{\lambda}, \qquad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^\top C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}, \quad (3)$$

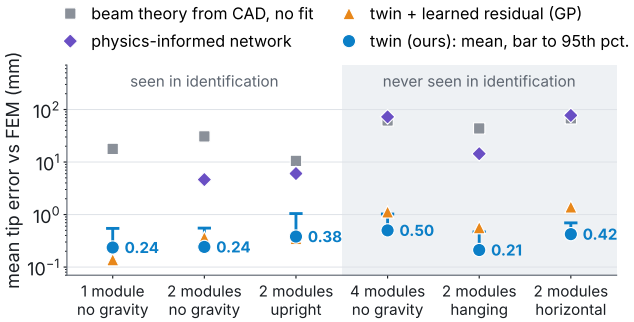
where $a_c = \int_c dy/A$ and $C_c = \int_c dy/J$, $\tau_{ic} = 1$ when tendon i passes slice c, and $c_t L_i$ is the stretch of the tendon. The map is exactly linear at any bend angle. Since a tendon cannot push, the tensions solve

$$\boldsymbol{\lambda}^\star = \arg\min_{\boldsymbol{\lambda}\geq\mathbf{0}} \tfrac{1}{2}\boldsymbol{\lambda}^\top G\,\boldsymbol{\lambda} - \delta\boldsymbol{\ell}^\top\boldsymbol{\lambda}, \quad (4)$$

a convex problem with a unique solution. Gravity in any direction is handled by a fixed-point iteration around this



(a) Twin (colour per module) against FEM (dots).



(b) Mean tip error on six held-out test sets.

Figure 8. Transfer to configurations never used for identification. The bars in (b) reach the 95th percentile of the twin's error; 60 to 100 poses per set.

Table 2. Mean tip error against FEM (mm).

Model	seen	new mounting	4 modules
Beam theory from CAD	10.5–30.8	43.6–68.0	61.7
Twin + GP residual	0.14–0.36	0.56–1.40	1.12
PINNs (forward, inverse)	3.4–6.0	9.2–77	54–73*
Identified law	**0.24–0.38**	**0.21–0.42**	**0.50**

*after retraining on four modules.

problem, and a stack is built by composing identical modules.

4.2 Learning the law of a module

Beam theory gets the structure right but underestimates how compliant a short notch is. Instead of training a network to output shapes, I fit the law of the module (Fig. 10). Each notch family k receives one scale,

$$C_\theta(y) = C_{\mathrm{CAD}}(y)\, e^{\theta_k}, \tag{5}$$

and θ is fitted through the exact solver to 675 SOFA poses of one and two modules. The notch compliances come out at 20.2 times the beam value axially and 5.9 and 5.3 times in bending, and a separate twist test gives a torsional factor of 2.45. These are physical properties of the module, so they carry over unchanged to any stack and mounting.

4.3 Results

Fig. 8 and Table 2 compare four models on held-out FEM poses. Beam theory from the CAD alone misses by 11 to 68 mm. A Gaussian-process residual on top of the identified twin is the most accurate on the configurations it was trained on but degrades on new ones. Forward and inverse PINNs trained on the same data fit their training sets to 3 to 6 mm, reach 9 to 77 mm in new mountings and have to be retrained for four modules. The identified law stays between 0.21 and 0.50 mm in every case, including four modules (640 mm) never used for fitting, and a pose takes 1 to 4 ms. The solver agrees with closed-form beam solutions to under 1 µm and with an independent Cosserat rod to 0.004 mm. All of these comparisons are against FEM; the hardware test is described in Section 5.

4.4 Preparing the hardware test

For the hardware build I designed the printable modules and couplers of Fig. 1b and computed a pretension ratio of 1 : 1.75 : 2.25 : 2.5 that keeps the distal tendons taut. Two identifiability results shape the experiment. Shape data cannot separate self-weight from stiffness, so the modules will be weighed rather than fitted. A 0.4 mm error in the tendon moment arm biases the law unless the moment arm is calibrated as a fifth parameter, which recovers it to 0.02 mm in simulation.

Table 3. Hardware risks and responses.

If the data show	then
residual growing with bend	replace the linear law by a convex notch energy $\psi_\theta(N, \boldsymbol{M})$ in the same solver
hysteresis	use the capstan friction term already in the code; load in one direction
mass traded against stiffness	weigh every module; never fit mass from shape
drift at a fixed command	fixed settling time; repeat poses at start and end

5 Future work

The two parts lead to one direction. SWEEP produces designs and the twin turns each design into a module law, so a law written as a function of geometry would tell us which backbone produces a given shape for a given stroke, in milliseconds. Fig. 9 shows the plan.

The first step is hardware validation with PhaseSpace motion capture. Five numbers are calibrated on about fifteen poses of two modules, and the twin is then tested on four modules, upright and hanging. The pass criterion was fixed before any data were taken: a mean tip error below 2 % of length, which is 13 mm for four modules. These results will complete the RA-L manuscript. The second step identifies the law for each SWEEP design and fits θ as a function of D_{RD}, D_i and L, first by regression and then with a neural operator [18] if the data call for one. It succeeds if it beats beam theory on designs held out entirely. The third step uses the model for inverse design and for closed-loop shape control. Table 3 lists what could go wrong on hardware and the planned response.

6 Conclusion

Part A showed that the notch geometry gives separate control of curvature and reach, at the cost of a stroke budget in which compression takes almost half. Part B showed that a module law identified inside an exact physical model transfers to longer stacks and new mountings, where learned solutions do not. Beyond the results, the semester taught me to check simulators and novelty early and to make every number reproducible.

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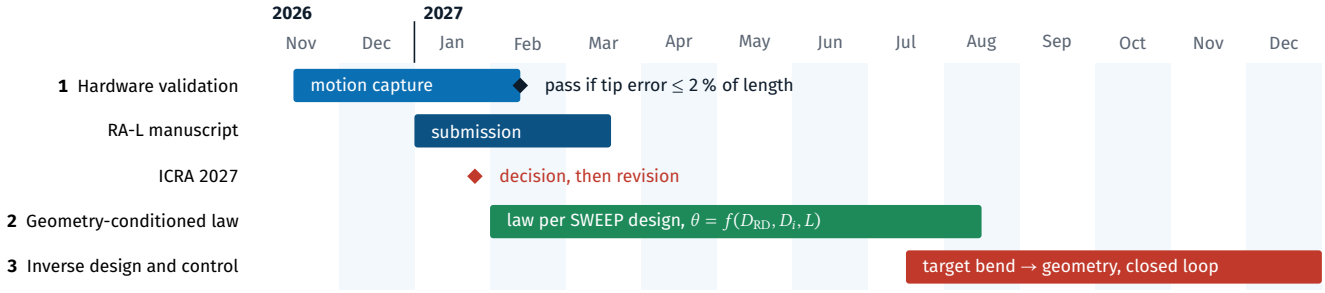


Figure 9. Planned schedule. Each step has a pass criterion fixed in advance; the dates are targets.

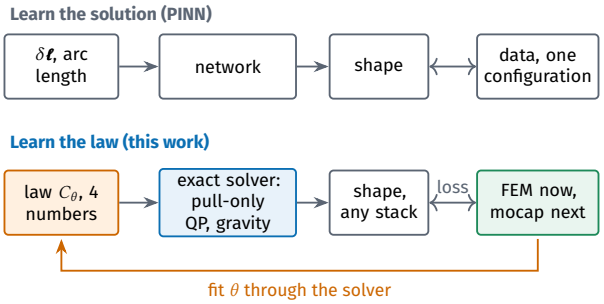


Figure 10. Two ways to bring learning into the model. A PINN learns the solution for one configuration and is retrained for the next; the identified law sits inside the solver and transfers by construction.

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